1.8.11 Validity Problem for Propositional and Predicate Logic

- We have seen that a formula (also called well formed formula or wff) of the propositional logic is an assertion involving propositional variables by using the connectives $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ in a proper manner.

- For example $p \wedge q$ is a wff of the propositional logic. When we assign particular propositions to p and q, we are giving an interpretation for the formula. If we say p stands for "I have exam tomorrow" and q stands for "I am going to study", $p \wedge q$ gets the interpretation 'I have exam tomorrow and I am going to study.'

- A wff of the propositional logic is a tautology if the wff always takes the value true whatever interpretation is given to it. $(p \vee \neg p)$ is always true whatever the proposition is assigned to p. Such a wff is called a tautology.

- Similarly a wff which is always false for all interpretations is called a contradiction.

- A wff which is neither a contradiction nor a tautology is called a contingency, i.e. for some interpretations it takes the value true and for other interpretations it takes the value False.

- The validity problem for the propositional logic is that given a wff of propositional logic, does there exist an interpretation which will make the wff take the value True.

- Considering the propositional variables as Boolean variables, this is called the Boolean Satisfiability Problem. This is a decidable problem in the sense we can give an algorithm to find out if the given wff is a contingency or tautology. The simplest way is to draw the truth table for the wff and if there is atleast one assignment for which the wff takes the value true, it is a contingency. If for all assignments it takes the value true, it is a tautology.

- In the first order logic, a predicate has n arguments. If P is a n-place predicate constant and values $c_1, c_2, c_3, \ldots, c_n$ are assigned to each of the individual variables, the result is a proposition.